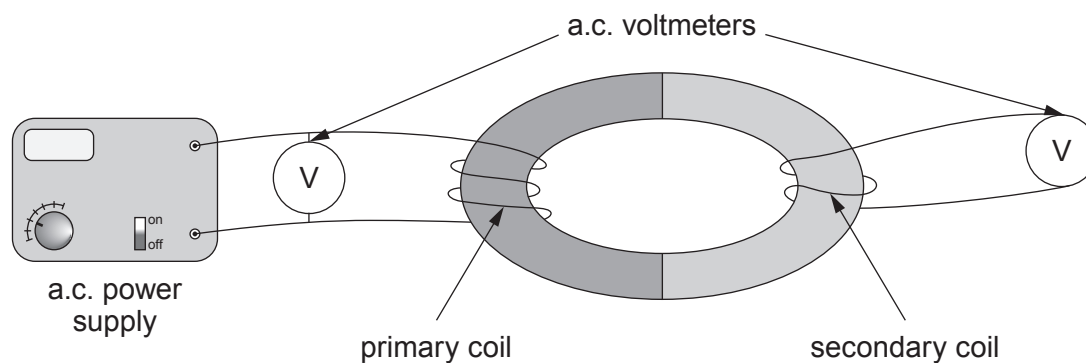


8. Transformers are used to change the size of an a.c. voltage.

- (a) Describe how the apparatus in the diagram below is used to investigate the effect of the number of turns on the secondary coil on the output of a transformer. [4]



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- (b) Explain why transformers only change a.c. voltages. [2]

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- (c) Explain how step-up transformers increase the efficiency of the National Grid system. [2]

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- (d) A transformer is used to reduce the 230V mains voltage to 11.5V to run a television. The primary coil of the transformer has 600 turns. Use an equation from page 2 to calculate the number of turns on the secondary coil. [3]

number of turns =

END OF PAPER

